MECH 325
Homework Assignment #3

Problem 1

For the following circuit, the relief pressure valve is set to 1000psi. The piston diameter is 4 inches and the rod diameter is 2 inches and the pump outputs a flow rate of 14 gpm. Determine:

a) The estimated input power required for the pump.
b) The maximum load for extension and retraction.
c) The speed of extension and retraction.
d) The flow rates in the lines and the required sizes for the piping system.

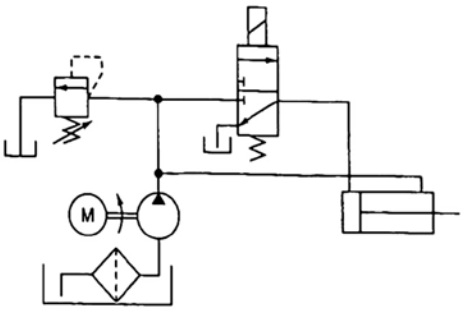


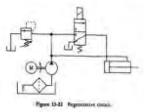
Figure 13-22 Regenerative circuit.

Problem 1

1— For the following circuit, the relief pressure valve is set to 1000psi. The piston diameter is 4 inches and the rod diameter is 2 inches and the pump outputs a flow rate of 14 gpm. Determine:

a) The estimated input power required for the pump.
b) The maximum load for extension and retraction.
c) The speed of extension and retraction.
d) The flow rates in the lines and the required sizes for the piping system.

$\rightsquigarrow \frac{gpm \times psi}{1714 \times 0.85} = \frac{14 \times 1000}{1714 \times 0.85} = 9.61\,hp \simeq 10\,hp$



given :
- relief pressure valve (working pressure) = 1000psi
- piston diameter (d_p) = 4in
- rod " (d_r) = 2in
- Q = 14 gpm

b)

Extension $\quad P = \frac{F}{A} \rightarrow F = P \times A$

$F_{Total} = F_p - F_r$

$= (P \times A_p) - (P \times A_r) = P(A_p - A_r)$

$= P\left[\frac{\pi}{4}D_p^2 - \frac{\pi}{4}(D_p^2 - D_r^2)\right] = P\left(\frac{\pi}{4}D_r^2\right) = (1000psi)\left(\frac{\pi}{4}\right)(2in)^2 = 3141\,lbs$

Retraction

$F_{Total} = F_r - F_p = P\left(\frac{\pi}{4}D_p^2 - \frac{\pi}{4}D_r^2\right) - P\left(\frac{\pi}{4}D_p^2\right)$

↳ in this case this $P = P_{atm} = 0$ gauge

$= P\frac{\pi}{4}(D_p^2 - D_r^2) = (1000psi)\left(\frac{\pi}{4}\right)(4in^2 - 2in^2) = 9424\,lbs$

c) $Q = V \times A$ (V: velocity, A: cross section area)

$Q_T = Q_{pump} + Q_{Regain}$

pump → (P), piston → (P), rod → (r)

- Regenerative → Extension (E)
- Regular → Retraction (R)

Extension

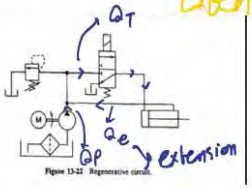


Figure 13-22 Regenerative circuit.

Extension

$Q_T = Q_P + Q_E \quad \therefore Q_P = Q_T - Q_E$

$Q_P = V_P\left(\frac{\pi}{4}D_P^2\right) - \left(V_P\frac{\pi}{4}(D_P^2 - D_r^2)\right)$

$\therefore Q_P = V_P\left(\frac{\pi}{4}D_r^2\right)$

$V_E = V_P = \frac{Q_{P\to pump}}{\left(\frac{\pi}{4}D_r^2\right)} = \frac{(21\text{gpm})(231\text{ in}^3)}{\frac{\pi}{4}(2\text{in}^2)} = 1029\text{ in/min}$ (piston)

$Q_T = (V_P)\left(\frac{\pi}{4}D_P^2\right) = \left(1029\frac{\text{in}}{\text{min}}\right)\left(\frac{\pi}{4}\times 4\text{in}^2\right) = 12.93\text{in}^3/\text{min}$

$Q_{regain} = Q_T - Q_P = 56\text{ gpm} - 14\text{ gpm} = 42\text{gpm}$

Retraction

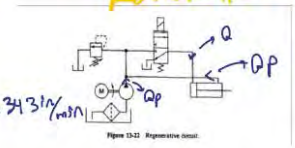


Figure 13-22 Regenerative circuit.

$Q_P = V_P \cdot (A_r)$ (pump, piston, retraction)

$\therefore V_R = V_P = \frac{Q_P}{A_r} = \frac{(14\text{gpm})(231\text{in}^3)}{\frac{\pi}{4}(4\text{in}^2 - 2\text{in}^2)} = 343\text{in/min}$ (rod)

$Q_{out} = (V_P)(A_P) = \left(343\frac{\text{in}}{\text{min}}\right)\left(\frac{\pi}{4}\times 1\text{in}^2\right) = \frac{430\text{ in}^2}{231}/\text{min} = 18.7\text{gpm}$ (or V_R)

Considering Q_T → to determine the sizes for piping system

$Q_{max} = Q_{Total\ extension} = 56$ gpm

$Q_T = (V_{max}) \times (\text{Area of pipe})$ & $V_{max} = 15$ ft/s → according to formula sheet

Curve always use $V_{max} = 15$ ft/s

∴ we should find minimum pipe size for this constraint)

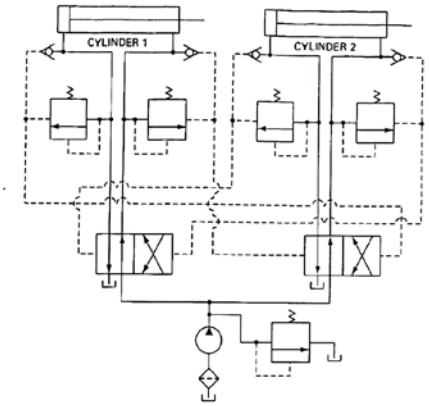
d) $\therefore \text{Area of pipe} = \frac{Q_T}{V_E} = \frac{(56\text{gpm})(231\text{in}^3/\text{gal})\left(\frac{1\text{m}}{60\text{s}}\right)}{(15\text{ ft/s})(12\text{ in/ft})} = 1.20\text{in}$

$d_P = \sqrt{\frac{4Q}{\pi V}\left(\frac{231}{720}\right)}$ (Q is in GPM, V is 15 ft/s)

→ round this up to nearest standard size from table to select pipe size

$\therefore$ pipe diameter $= 1.23''$ → select $1\frac{1}{4}''$ 40 pipe

Problem 2



For the above hydraulic circuit, give the sequence of operation of the cylinders when the pump is turned on. Assume that the cylinders are both fully retracted at the start. The directional control values are controlled by hydraulic solenoids activated by pressurized pilot lines.

1. Both cylinders are fully retracted, and high-pressure fluid flows into the rod side, keeping them retracted initially. Pressure builds up on the right-side pressure control valves of both cylinders. Eventually, the pressure is high enough to flow through the valve, into the directional valve's control lines.

2. Assuming that a pressure build up on one side of the directional valve pushes that valve along the pressure gradient, the directional control valve of Cylinder 1 moves to the next position, while the directional control valve of Cylinder 2 remains in the original position.

3. This allows high pressure fluid from the pump to flow to the piston end of Cylinder 1, therefore causing cylinder 1's extension.

4. After cylinder 1 finishes extending all the way, pressure continues to build up. Fluid flows towards cylinder 2's directional control valve, and it shifts position to the right-side configuration.

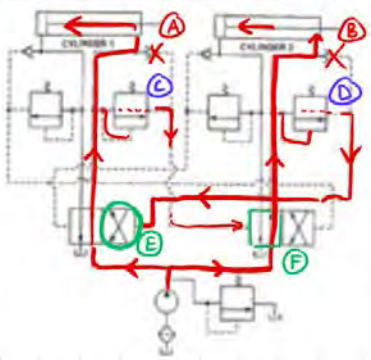
5. The high-pressure fluid from the pump flows to the piston side of cylinder 2, causing extension of cylinder 2. The high-pressure fluid then flows to the directional control valve of cylinder 1, causing a shift to the original side of the directional control valve.

6. High pressure fluid from the pump then enters the rod side of cylinder 1, causing the piston to retract.

7. The high pressure builds and passes through the right-side flow control valve of cylinder 1, causing the directional control valve of cylinder 2 to shift to its original position again.

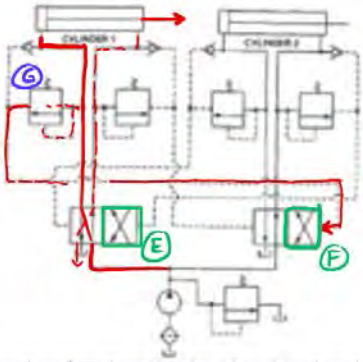
8. High pressure fluid from the pump then enters the rod side of cylinder 2, causing the piston to retract. The process repeats from the initial position again.

1)



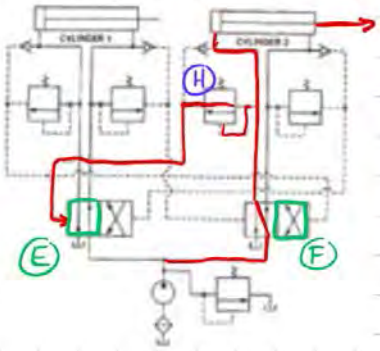
- both cylinders stay fully retracted
- check valves closed (A + B)
- pressure control valves C + D open
- hydraulic solenoid E activated, switching direction control valve

2)



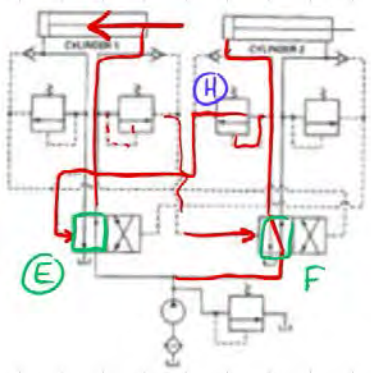
- cylinder 1 extends
- once fully extended, pressure builds up to open valve G, actuating hydraulic solenoid at F and switching direction control valve

3)

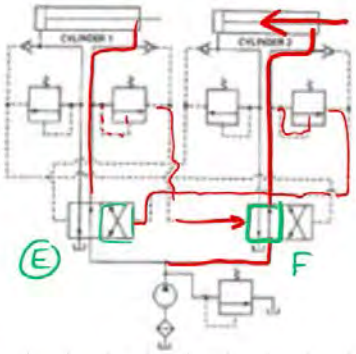


- cylinder 2 extends
- once fully extended, pressure builds and opens pressure valve H, causing direction control valve E to return to original state

4)



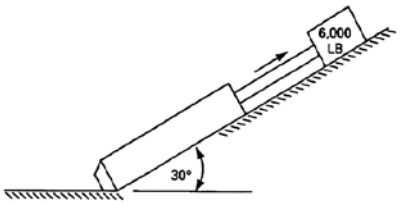
- cylinder 1 retracts, then switches valve F back to original state



- cylinder 2 retracts, then pressure builds and flips direction control valve E, and cycle repeats.

1 EXTEND, 2 EXTEND, 1 RETRACT, 2 RETRACT, (REPEAT)

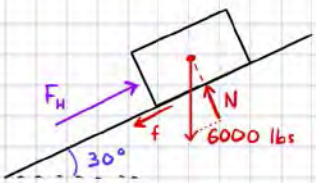
Problem 3



A hydraulic cylinder is required to push a 6000 pound load up a 30° ramp. Assume the cylinder is rigidly attached to the surface but the load is not guided. The stroke length is 20 inches and relief valve is set to 1000psi. The coefficient of friction for the load and ramp is 0.2.

a) Determine the required cylinder bore and rod diameter.
b) If the load must be accelerated to 0.5 ft/sec in 0.5 inches, will your cylinder bore is sufficient?

a)



$+\rightarrow \sum F = ma$

$F_H - 6000 \sin 30° - f = ma$

$f = \mu N = 0.2\,(6000 \cos 30°) = 1200 \cos 30°$

$= 1039.23$ lb

$F_H > 6000 \sin 30° + 1200 \cos 30°$ to accelerate load

$\boxed{F_H > 4039.23 \text{ lb}}$

$P = \frac{F}{A} \rightarrow A = \frac{F}{P} = \frac{4039.23 \text{ lb}}{1000 \text{ psi}} = 4.03923 \text{ in}^2 = \frac{\pi d^2}{4}$

$d_{min} = 2.27$ in minimum - round up to standard size

$\rightarrow$ choose $d = 2.5$ in bore

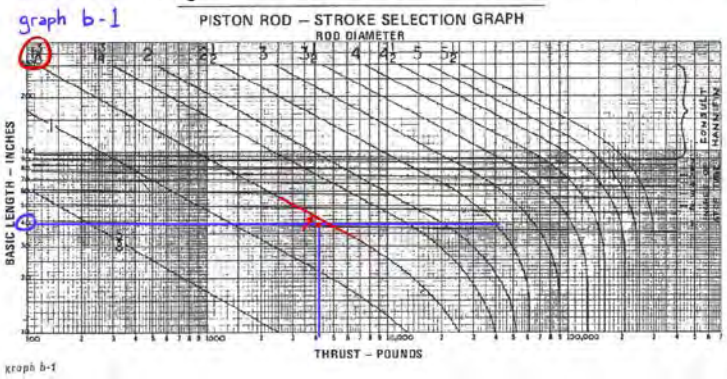
ROD DIAMETER :

Parker, Table b-4

- supported, not rigidly guided → STROKE FACTOR = 2.0

Basic Length = 20 in × 2.0 = 40 in

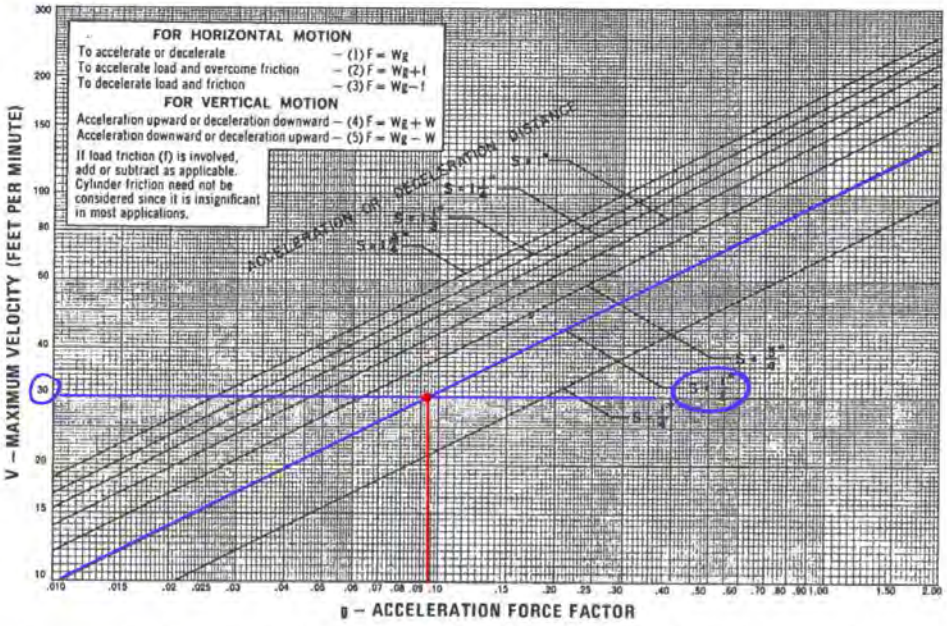
graph b-1



Choose $1\,3/8$" ROD

b) accelerate to 0.5 ft/s in 0.5 in
(6 in/s)

graph b-2



$V_{max} = 0.5\ ft/s \times \frac{60s}{min.} = 30\ ft/min$

S = acceleration distance = 0.5 in

→ g = 0.095

acceleration force = 6000 lb (0.095) = 570 lb

friction = 1039.23 lb

weight = 6000 sin 30° = 3000 lb

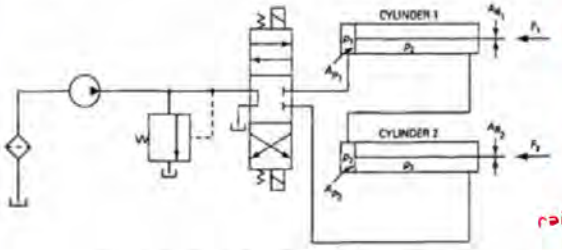
F_{TOTAL} = 4609.23 lb required

$F = P \cdot A = P \cdot \frac{\pi d^2}{4}$ @ d = 2.5" $F = 1000\ psi \left(\frac{\pi \cdot 2.5^2}{4}\right) in^2 =$ 4908.7 lb

✓ OK

2.5 in bore is sufficient

4) **Problem 4**



3000 lb

1000 lb

raise to 40"

Figure 9-13. Cylinders hooked in series will operate in synchronization.

The above circuit is used to raise a platform rated to a maximum load of 4000 lbs. The distribution of the load is such that cylinder 1 will lift 3000 lbs while cylinder two will lift 1000 lbs. The platform must be raised a maximum of 40 inches.

a) If the working pressure is 1000psi, size the required cylinder bores, rod diameters, and pump capacity if we wish to raise the platform at 15 ft/min. You can assume that the cylinders are rigidly mounted and that the platform is fixed and rigidly guided.
b) If the platform must be accelerated to the final velocity within ¾", will the cylinder bores be sufficient?

Table b-4:

Fixed + Rigidly guided:

Stroke Factor = 0.50

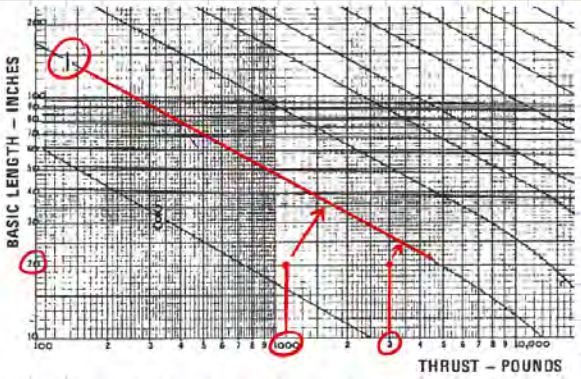
Basic Length = 40" × 0.50 = 20"

a) outflow from cylinder 1 = inflow to cylinder 2 P_1 = 1000 psi working pressure

ROD DIAMETER (Graph b-1)

@ 3000 lb, 20 in basic length → 1 in rod required for cylinder 1

@ 1000 lb, 20 in B.L. → 1 in rod required for cylinder 2



(minimum rod size - enlarge if needed for appropriate flow rate)

FIND PAIR OF STANDARD SIZED CYLINDERS WITH SIMILAR OR EQUAL OUTFLOW (1) TO INFLOW (2) - RATE OF EXTENSION MUST BE IDENTICAL!

Parker [Table b-5]: Choose 5" BORE, 3" ROD FOR (1) } BOTH REQ. 6.53 GPM @ 10 ft/s

4" BORE, 1" ROD FOR (2)

= 9.795 GPM @ 15 ft/s

(Note: Cylinders are overspecced, so working pressure will likely need to be reduced by adjusting relief valve)

$$A_1 = \frac{\pi(5^2)}{4} = 19.635 \text{ in}^2$$

$$A_2 = \frac{\pi(4)^2}{4} = A_1 - A_{ROD1} = \frac{\pi(5^2 - 3^2)}{4}$$

$$A_2 = 12.566 \text{ in}^2$$

$F_{1\,MAX} = 1000\left(\frac{\pi \cdot 5^2}{4}\right) = 19\,635\ lb$ (much higher than required force ~ 4000 lb)

$F_{2\,MAX} = F_{1\,MAX} - 3000\ lb \approx 16\,634\ lb$

REDUCE PRESSURE → $P_2 A_2 = 1000\ lb$

$P_2 = 79.6\ psi$ required

$P_1 A_1 = P_2(\underbrace{A_1 - A_{ROD}}_{A_2}) + 3000$ $\quad P_1 = \frac{4000}{A_1} = 203.7\ psi$ required working pressure

PUMP CAPACITY:

5" cylinder @ 10 fpm requires 10.2 GPM

$\frac{10.2\ GPM}{10\ fpm} \times 15\ fpm = 15.3\ GPM$ capacity required

CYLINDER 1: 5 in BORE, 3 in ROD

CYLINDER 2: 4 in BORE, 1 in ROD

PUMP: 15.3 GPM required

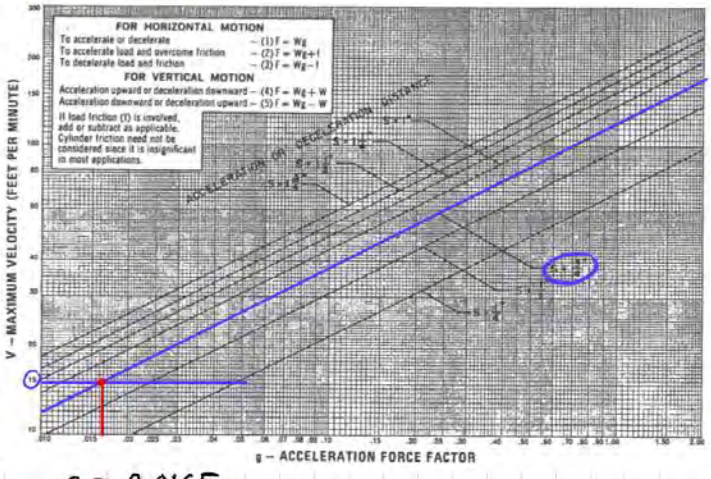
(working pressure reduced to ≈ 204 psi)

* CYLINDERS are significantly larger than what is required to lift the load, but the geometry is necessary to provide identical extension rates (to keep the platform level)

- if some amount of discrepancy / "crookedness" was acceptable (not specified in question), then smaller cylinders could be used instead @ higher working pressure

b) $s = 3/4"$, $V_{max} = 15\ fpm$ - Check acceleration

[Graph b-2]:



- assume vertical motion ("lifting")

g = 0.0165

$F = W_g + W$

$= 3000\,(0.0165) + 3000 + 1016.5$ ← $= 1000\,(0.0165) + 1000$

≈ 4066 lb required for ① ✓ OK

$= 1016.5$ lb req. for ② ✓ OK

CYLINDERS CHOSEN IN PART a) ARE SUFFICIENT

- adjust working pressure to provide required force or greater

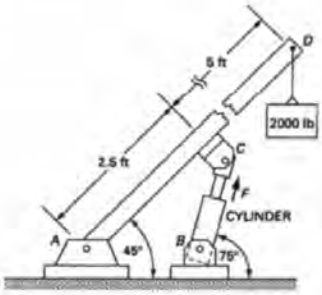
$P_1 A_1 \geq 4066$ lb

$P_1 \geq \dfrac{4066 \text{ lb}}{\left(\frac{\pi \cdot 5^2}{4}\right) \text{in}^2}$

$P_1 \geq 207.1$ psi working pressure

(- higher pressure will accelerate loads more quickly / in less distance)

5) **Problem 5**



a) Determine the required force, F, to support the 2000 lb. load in the position shown above.

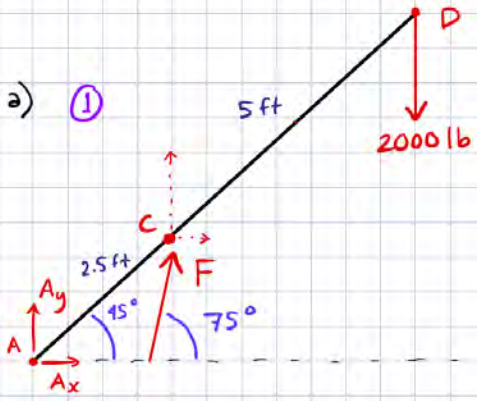
b) With a working pressure of 1000psi, select an appropriate cylinder size for system (for the position shown).

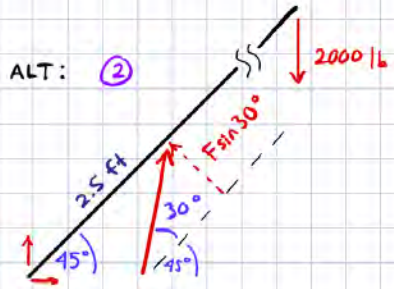
c) If the extension for the cylinder is 14 inches, determine the required rod diameter for the system (for the position shown).

d) If we wish to lift the load at a rate of 8 inches/sec (at the position shown), what size of pump, piping system, and electric motor are required?

e) If we include the fact that we wish to accelerate the cylinder to its required speed within ¼", will this change the results above (at the position shown)?

f) Looking at the system, do you expect the required thrust to be higher at the starting position (fully retracted rod) or at the fully extended position? Let's assume the position above is for the 50% point.





a) ①

$$+\circlearrowleft \Sigma M_A = 0$$

$$-2000(7.5\sin 45^\circ) - F\cos 75^\circ (2.5\sin 45^\circ) + F\sin 75^\circ (2.5\sin 45^\circ) = 0$$

$$F\left(2.5\frac{\sqrt{2}}{2}\right)(\sin 75^\circ - \cos 75^\circ) = 15000\frac{\sqrt{2}}{2} = 10606.602 \text{ lb-ft}$$

$$F = \frac{15000}{2.5(\sin 75^\circ - \cos 75^\circ)} = 8485.28 \text{ lb}$$

$$\boxed{F = 8485.3 \text{ lb}}$$

(② Check: $+ F\sin 30^\circ (2.5) - 2000\sin 45^\circ (7.5) = 0$

$$F = \frac{2000(7.5)\sin 45^\circ}{2.5\sin 30^\circ} = 8485.3 \checkmark$$)

b) $P = 1000$ psi

$$A = \frac{F}{P} = \frac{8485.3 \text{ lb}}{1000 \text{ lb/in}^2} = 8.4853 \text{ in}^2 \text{ required (for push)}$$

(minimum)

$$A = \frac{\pi d^2}{4} \rightarrow d = \left(\frac{4A}{\pi}\right)^{1/2} = \sqrt{\frac{4 \cdot 8.4853 \text{ in}^2}{\pi}}$$

$d_{min} = 3.287$ in → round up to 3.5 in (standard size)

not in Table 6-5?

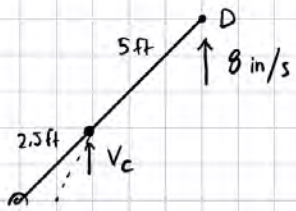
$\boxed{d = 3.5 \text{ in}}$ ($F_{max} = 9620$ lb) // OR $\boxed{d = 4 \text{ in}}$ ($F_{max} = 12570$ lb) - see Table 6-5

c) Extension : 14 inches → check buckling

Table b-4 : Pivoted + Rigidly guided : $k = 2$ → $B.L. = 14 \cdot 2.0 = 28"$

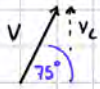
Graph b-1 : @ ~8500 lb, 28" B.L → **ROD DIAMETER = 1 3/8"**

d) Lift Load @ 8 in/sec



$V_c = 8 \cdot \frac{2.5}{7.5} = 2.667$ in/s $\times \frac{60s}{min} \times \frac{1ft}{12in}$

$= 13.33$ ft/min. ↑



$V \sin 75° = 13.33$ fpm

$V = 13.804$ fpm extension

[Table b-5] : @ 10 fpm, 6.53 GPM req.

PUMP : $Q = \frac{6.53 \text{ GPM}}{10 \text{ fpm}} \times 13.804 \text{ fpm} = 8.76554$ GPM

Q = 8.77 GPM

PIPE : TRY 1/2" : $\frac{8.95 \text{ fpm}}{6.53 \text{ GPM}} \times 8.77 \text{ GPM} = 12.02$ fpm < 15 fpm ✓ OK

PIPE = 1/2 in

MOTOR : [Table c-1] : @ 1000 psi, ~ 9 GPM : 6.18 hp required

From [Table c-3] - round up to 7.5 hp 220 V or 440 V
22 A 11 A

e) accelerate within 3/4"

$S = 3/4"$, $V_{max} \approx 13.8$ fpm [Graph b-2] : $g = 0.014$ for 2000 lb

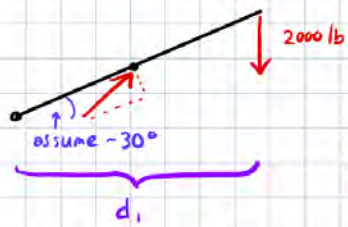
$= 28$ lb extra

$28 \times \frac{0485}{2000} = 119$ lb extra @ actuator (point C) $= 8604$ lb total

→ still below $F_{max} = 12570$ lb for $d = 4$ in, so no change required

f)

STARTING vs. EXTENDED



d_1

$$F \approx \frac{2000\,(7.5\cos 30^\circ)}{2.5\sin 30^\circ} \approx 10\,400\text{ lb}$$



d_2

$$F \approx 2000\left(\frac{7.5}{2.5}\right)\cdot\cos 60^\circ \approx 3000\text{ lb}$$

Expect higher thrust required at **STARTING POSITION**

- moment arm for load is longer at starting position ($d_1 > d_2$), so actuator must provide higher force to balance moments and hold equilibrium
- actuator angle is shallower @ start, so less force is perpendicular to the arm - a greater portion is "wasted" pushing against the hinge